Qualcomm Technologies, Inc.

DragonBoard™ 410c based on Qualcomm® Snapdragon™ 410 processor

GPIO Pin Assignment Spreadsheet

May 20, 2015

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Revision history

Revision	Date	Description
B	May 2015	URL removed from cover
A	May 2015	Initial release

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1 System I/O

This document provides information on system General Purpose Input/Outputs (GPIOs) for DragonBoard™ 410c based on Qualcomm® Snapdragon™ 410 processor (APQ8016). Use this document with the DragonBoard 410c schematics.

Table 1-1 System GPIOs for DragonBoard 410c

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	DragonBoard 410c
GPIO_0		3	1.8 V	PD	LS EXP CONN UART0 (GP)	blsp_uart_tx[1]
GPIO_1	Y	3	1.8 V	PD		blsp_uart_rx[1]
GPIO_2		3	1.8 V	PD		blsp_uart_cts_n[1]
GPIO_3		3	1.8 V	PD		blsp_uart_rfr_n[1]
GPIO_4		3	1.8 V	PD	LS EXP CONN UART1 (debug)	blsp_uart_tx[2]
GPIO_5	Y	3	1.8 V	PD		blsp_uart_rx[2]
GPIO_6		3	1.8 V	PD	LS EXP CONN I2C0 (sensors)	blsp_i2c_sda[2]
GPIO_7		3	1.8 V	PD		blsp_i2c_scl[2]
GPIO_8		3	1.8 V	PD	HS EXP CONN SPI1	blsp_spi_mosi[3]
GPIO_9	Y	3	1.8 V	PD		blsp_spi_miso[3]
GPIO_10		3	1.8 V	PD		blsp_spi_cs_n[3]
GPIO_11	Y	3	1.8 V	PD		blsp_spi_clk[3]
GPIO_12	Y	3	1.8 V	PD	LS EXP CONN - GPIOs (touch screen)	GPIO-B (TS_RST_N)
GPIO_13	Y	3	1.8 V	PD		GPIO-C (TS_INT_N)
GPIO_14		3	1.8 V	PD	HS EXP CONN (I2C3) DSI2HDMI	blsp_i2c_sda[4]
GPIO_15		3	1.8 V	PD		blsp_i2c_scl[4]
GPIO_16		3	1.8 V	PD	LS EXP CONN SPI0 (touch screen)	blsp_spi_mosi[5]
GPIO_17		3	1.8 V	PD		blsp_spi_miso[5]
GPIO_18		3	1.8 V	PD		blsp_spi_cs_n[5]
GPIO_19		3	1.8 V	PD		blsp_spi_clk[5]
GPIO_20	Y	3	1.8 V	PD	Display	HDMI_HPD_N
GPIO_21	Y	3	1.8 V	PD	LEDs	USR_LED_1_CTRL
GPIO_22		3	1.8 V	PD	LS EXP CONN I2C1	blsp_i2c_sda[6]
GPIO_23		3	1.8 V	PD		blsp_i2c_scl[6]

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	**DragonBoard 410c**
GPIO_24		3	1.8 V	PD	LS EXP CONN – GPIOs (display)	GPIO-G (DSI_VSYNC)
GPIO_25	Y	3	1.8 V	PD		GPIO-H (DSI_RST)
GPIO_26		3	1.8 V	PD	HS EXP CONN – CAM MCLK	CSI0_MCLK
GPIO_27		3	1.8 V	PD		CSI1_MCLK
GPIO_28	Y	3	1.8 V	PD	LS EXP CONN – GPIOs (camera)	GPIO-K (CSI1_RST)
GPIO_29		3	1.8 V	PD	HS EXP CONN I2C2 (CCI)	cci_i2c_sda0
GPIO_30		3	1.8 V	PD		cci_i2c_scl0
GPIO_31	Y	3	1.8 V	PD	Display	DSI2HDMI_INT_N
GPIO_32		3	1.8 V	PD	Display	DSI_SW_SEL
GPIO_33		3	1.8 V	PD	EXP CONN – GPIOs (camera)	GPIO-L (CSI1_PWDN)
GPIO_34	Y	3	1.8 V	PD		GPIO-J (CSI0_PWDN)
GPIO_35	Y	3	1.8 V	PD		GPIO-I (CSI0_RST)
GPIO_36	Y	3	1.8 V	PD	LS EXP CONN – GPIOs	GPIO_A (APQ_INT)
GPIO_37	Y	3	1.8 V	PD	For factory emergency download	forced_usb_boot
GPIO_38	Y	3	1.8 V	PD	SD card	#sd_card_det_n#

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	**DragonBoard 410c**
GPIO_39		3	1.8 V	PD	WLAN, BT, FM	wcss_bt_ssbi
GPIO_40		3	1.8 V	PD		wcss_wlan_data[2]
GPIO_41		3	1.8 V	PD		wcss_wlan_data[1]
GPIO_42		3	1.8 V	PD		wcss_wlan_data[0]
GPIO_43		3	1.8 V	PD		wcss_wlan_set
GPIO_44		3	1.8 V	PD		wcss_wlan_clk
GPIO_45		3	1.8 V	PD		wcss_fm_ssbi
GPIO_46		3	1.8 V	PD		wcss_fm_sdi
GPIO_47		3	1.8 V	PD		wcss_bt_ctl
GPIO_48		3	1.8 V	PD		wcss_bt_dat_stb
GPIO_49	Y	4	1.8 V/2.85 V	PD		NC
GPIO_50	Y	4	1.8 V/2.85 V	PD		NC
GPIO_51	Y	4	1.8 V/2.85 V	PD		NC
GPIO_52	Y	3	1.8 V	PD		NC
GPIO_53		6	1.8 V/2.85 V	PD		NC
GPIO_54	Y	6	1.8 V/2.85 V	PD		NC
GPIO_55		6	1.8 V/2.85 V	PD		NC
GPIO_56		3	1.8 V	PD		NC
GPIO_57		5	1.8 V/2.85 V	PD		NC
GPIO_58		5	1.8 V/2.85 V	PD		NC
GPIO_59		5	1.8 V/2.85 V	PD		NC
GPIO_60		3	1.8 V	PD		NC
GPIO_61		3	1.8 V	PD		NC
GPIO_62	Y	3	1.8 V	PD		NC

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	**DragonBoard 410c**
GPIO_63		3	1.8 V	PD	Audio PDM	CDC_PDM0_CLK
GPIO_64		3	1.8 V	PD		CDC_PDM0_SYNC
GPIO_65		3	1.8 V	PD		CDC_PDM0_TX0
GPIO_66	Y	3	1.8 V	PD		CDC_PDM0_RX0
GPIO_67		3	1.8 V	PD		CDC_PDM0_RX1
GPIO_68	Y	3	1.8 V	PD		CDC_PDM0_RX2
GPIO_69	Y	3	1.8 V	PD	LS EXP CONN – GPIOs (sensors + APQ wake)	GPIO_D (MAG_INT)
GPIO_70		3	1.8 V	PD		NC
GPIO_71		3	1.8 V	PD		NC
GPIO_72		3	1.8 V	PD		NC
GPIO_73		3	1.8 V	PD		NC
GPIO_74		3	1.8 V	PD		NC
GPIO_75		3	1.8 V	PD		NC
GPIO_76		3	1.8 V	PD		NC
GPIO_77		3	1.8 V	PD		NC
GPIO_78		3	1.8 V	PD		NC
GPIO_79		3	1.8 V	PD		NC
GPIO_80		3	1.8 V	PD	BOOT CONFIG	boot_config[0] (WDOG_DIS)
GPIO_81		3	1.8 V	PD		boot_config[1]
GPIO_82		3	1.8 V	PD		boot_config[2]
GPIO_83		3	1.8 V	PD		boot_config[3]
GPIO_84		3	1.8 V	PD		NC
GPIO_85		3	1.8 V	PD		NC
GPIO_86		3	1.8 V	PD	BOOT CONFIG	boot_config[5] (apps_boot_from_rom)
GPIO_87		3	1.8 V	PD		NC
GPIO_88		3	1.8 V	PD		NC

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	DragonBoard 410c
GPIO_89		3	1.8 V	PD		NC
GPIO_90		3	1.8 V	PD		NC
GPIO_91		3	1.8 V	PD		NC
GPIO_92		3	1.8 V	PD		NC
GPIO_93		3	1.8 V	PD		NC
GPIO_94		3	1.8 V	PD		NC
GPIO_95		3	1.8 V	PD		NC
GPIO_96		3	1.8 V	PD		NC
GPIO_97	Y	3	1.8 V	PD		NC
GPIO_98	Y	3	1.8 V	PD		NC
GPIO_99		3	1.8 V	PD		NC
GPIO_100		3	1.8 V	PD		NC
GPIO_101		3	1.8 V	PD		NC
GPIO_102		3	1.8 V	PD		NC
GPIO_103		3	1.8 V	PD		NC
GPIO_104		3	1.8 V	PD	GPS control	SSBI_GPS
GPIO_105		3	1.8 V	PD		NC
GPIO_106		3	1.8 V	PD		NC
GPIO_107	Y	3	1.8 V	PD	Keypad	#KEY_VOL_UP_N#
GPIO_108	Y	3	1.8 V	PD		NC
GPIO_109	Y	3	1.8 V	PD		NC
GPIO_110	Y	3	1.8 V	PD	LS EXP CONN (I2S)	pri_mi2s_ws
GPIO_111	Y	3	1.8 V	PD		NC
GPIO_112	Y	3	1.8 V	PD		NC
GPIO_113	Y	3	1.8 V	PD	LS EXP CONN (I2S)	pri_mi2s_sck (ALSP_INT)
GPIO_114	Y	3	1.8 V	PD		pri_mi2s_data0

GPIO#	MPM IRQ	Pad group	Pad voltage	Dft pull	DragonBoard 410c	
					Description	**DragonBoard 410c**
GPIO_115	Y	3	1.8 V	PD	LS EXP CONN – GPIOs (sensors)	GPIO_E (GYRO_ACCL_INT_N)
GPIO_116		3	1.8 V	PD		
GPIO_117	Y	3	1.8 V	PD	DSI2HDMI – I2S	sec_mi2s_ws
GPIO_118	Y	3	1.8 V	PD		sec_mi2s_sck
GPIO_119		3	1.8 V	PD		sec_mi2s_data0
GPIO_120	Y	3	1.8 V	PD	LEDs	USR_LED_2_CTRL
GPIO_121	Y	3	1.8 V	PD	USB	USB_HS_ID
PM_GPIO_01					USB	USB_HUB_RESET_N
PM_GPIO_02					USB	USB_SW_SEL
PM_GPIO_03					LEDs	USR_LED_3_CTRL
PM_GPIO_04					LEDs	USR_LED_4_CTRL
MPP01						VDDPX_BIAS
MPP02					LEDs	WLAN_LED_SINK
MPP03					LEDs	BT_LED_SINK
MPP04					LS EXP CONN – GPIOs (display)	GPIO_F (DSI_BLCTL)

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